Q1.

Figure 1 shows a pool full of water.

(Source: © AnaKulagina/Shutterstock)

Figure 1

(i) The pressure on the ground due to the water in the pool is 3500 Pa.

The density of water is 10^3 kg/m^3.

The gravitational field strength, *g*, = 10 N/kg.

Calculate the depth of the water in the pool.

(2)

depth = ... m

(ii) Figure 2 shows part of the side and base of the pool.

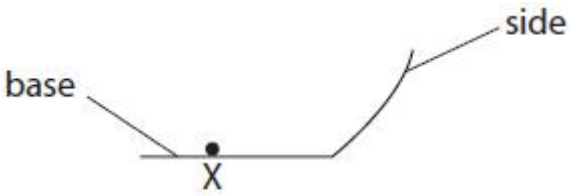


Figure 2

The water in the pool causes a force on the base of the pool.

Draw an arrow on Figure 2 to show the direction of this force at point X.

(1)

(Total for question = 3 marks)

Examiner's Report

Q1.

(i)
Most candidates used the correct equation and rearranged it to find h.

(i) The pressure on the ground due to the water in the pool is 3500 Pa.

The density of water is 10^3 kg / m^3.

The gravitational field strength, g, = 10 N / kg.

Calculate the depth of the water in the pool.

(2)

Pressure = h x p x g

3500 = h x (10^3) x 10

h = 3500 ÷ 10,000

depth = 0·35 m

Results Plus: Examiner Comments
A well laid out and fully correct answer. It scores the full two marks.

The most common error was to use 10 for the density of water rather than 10^3.

(i) The pressure on the ground due to the water in the pool is 3500 Pa.

The density of water is 10^3 kg / m^3.

The gravitational field strength, g, = 10 N / kg.

Calculate the depth of the water in the pool.

(2)

P = h x p x g

$\frac{P}{p \times g}$ = h $\frac{3500}{10 \times 10}$ = 35

depth = 35 m

Results Plus: Examiner Comments
Examiners would allow one mark for an answer of 35 m even though this is the height of a large building. It scored one out of two marks.

Results Plus: Examiner Tip
Hint: Look at the result of a calculation to see if the value makes sense.

(ii)
Candidates almost always drew an acceptable arrow pointing downwards either from or to the point X.

(ii) Figure 2 shows part of the side and base of the pool.

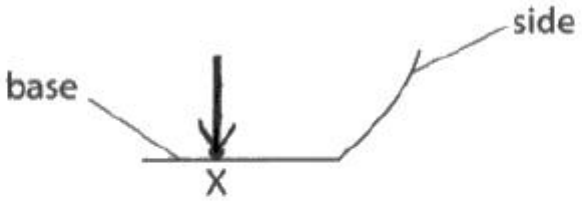


Figure 2

The water in the pool causes a force on the base of the pool.

Draw an arrow on Figure 2 to show the direction of this force at point X.

(1)

Results Plus: Examiner Comments
A clear and well drawn arrow in the correct position and pointing down. It scored the one mark for this question.

The most common error was to draw the arrow too far away from the point.

(ii) Figure 2 shows part of the side and base of the pool.

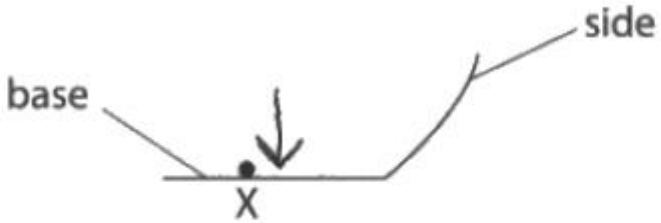


Figure 2

The water in the pool causes a force on the base of the pool.

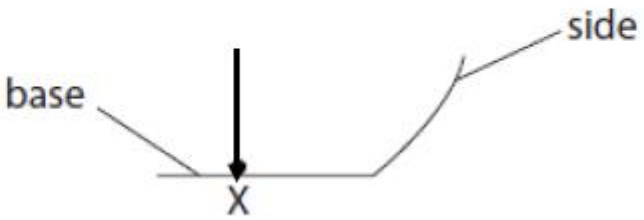
Draw an arrow on Figure 2 to show the direction of this force at point X.

Results Plus: Examiner Comments
An arrow like this does not show the force at X but somewhere in the middle of the pool. It scored zero marks.

Mark Scheme

Q1.

Question number	Answer	Additional guidance	Mark
(i)	substitution into $P = h \times \rho \times g$ (1) $3500 = h \times 10^3 \times 10$ rearrangement and evaluation (1) (h =) 0.35 (m) **OR ALTERNATIVE METHOD** rearrangement (1) $h = \frac{P}{\rho \times g}$ substitution and evaluation (1) (h =) 0.35 (m)	35 cm 35 cm accept 35 (m) for 1 mark award full marks for correct answer without working	**(2)** **AO2.1**

Question number	Answer	Additional guidance	Mark
(ii)	A straight, vertical, downwards arrow as shown  	Judge by eye. Arrow tip or tail (extended if necessary) must pass through point X.	**(1)** **AO1.1**